

PROJECT CHARTER

GENERAL PROJECT INFORMATION

PROJECT NAME

PROJECT MANAGER

PROJECT GUIDE

Generic Digital Signing Online Tool (GDSOT)

Deputy Director (IT) –
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National Informatics Centre (NIC)

PROJECT DURATION

SIX MONTH

EXPECTED START DATE

EXPECTED COMPLETION
DATE

NO BUDGET INVOLVED -INTERNSHIP-BASED-PROJECT

01/01/2026

31/06/2026

PROJECT OVERVIEW

PROBLEM OR ISSUE

Government operations depend heavily on the exchange of structured data across databases and systems. While relational database systems (such as PostgreSQL, MySQL, Oracle, and SQL Server) support efficient data storage and retrieval, there is no standardized mechanism to ensure that data remains authentic and unaltered once it is extracted or shared outside its original environment.

Without backend cryptographic validation, data workflows may face risks such as unauthorized modification, integrity compromise during transfer, and difficulty verifying authenticity. This can reduce trust in shared datasets. A secure and standardized solution is therefore required to ensure that imported and exported data remains tamper-evident and verifiable.

PURPOSE OF PROJECT

The purpose of this project is to develop a secure web-based integrity validation system capable of importing structured data from heterogeneous relational database systems through standardized connectivity interfaces. While the initial implementation and testing phase utilizes available PostgreSQL infrastructure within the NIC environment, the system architecture is designed to be database-agnostic and extensible.

The platform applies backend digital signature mechanisms to ensure cryptographic integrity and system-level authenticity of datasets prior to export. The system will enable export of digitally validated datasets in formats such as Excel and JSON, ensuring that transferred data remains verifiable and tamper-evident within a government operational context.

BUSINESS CASE	Government systems require secure and reliable mechanisms to ensure the integrity and authenticity of structured data exchanged across databases and operational units. The implementation of a web-based data import and digital signature system strengthens data governance by introducing cryptographic validation into backend processing workflows. By ensuring that imported and exported datasets are verifiable and tamper-evident, the system enhances operational trust, reduces integrity-related risks, and supports secure data handling practices within a government environment. As an internship-level implementation, the project provides a low-risk demonstration of scalable data integrity validation mechanisms suitable for broader adoption
GOALS / METRICS	• Secure import of structured data from supported relational database systems • Accurate transformation and local storage • Correct implementation of backend digital signature logic • Detection of tampering through integrity verification • Successful export of integrity-protected datasets (Excel and JSON) • Successful demonstration within NIC use case environment
EXPECTED DELIVERABLES	The project will deliver the following components: • A functional web-based data import, export and integrity validation system • Data import module • Local database processing and transformation module • Backend digital signature implementation for data integrity assurance • Data integrity verification mechanism • Export functionality supporting Excel, JSON,etc • Technical documentation and system architecture overview • Final system demonstration within NIC use-case context

PROJECT SCOPE

WITHIN SCOPE	• Import of structured data from any databases • Transformation and storage within a controlled local database • Backend digital signature application for integrity validation • Verification of data integrity through cryptographic validation • Export of digitally validated datasets in Excel, JSON, etc formats • Internship-level implementation under NIC supervision
OUTSIDE OF SCOPE	• Support for non-relational (NoSQL) databases in the initial phase • Real-time distributed data synchronization • Production-scale deployment and enterprise rollout • Advanced analytics or reporting dashboards • Biometric authentication or identity management modules

TENTATIVE SCHEDULE

KEY MILESTONE	START	FINISH
Form Project Team / Preliminary Review / Scope	MONTH 1	MONTH 1
Finalize Project Plan / Charter / Kick Off		
Define Phase		
Measurement Phase		
Analysis Phase		
Improvement Phase		
Control Phase		
Project Summary Report and Close Out		

RESOURCES

PROJECT TEAM	PROJECT GUIDE/MENTOR Rajeev Rai — Deputy Director (IT) Project guidance, review, and approval Core Project Team Members Rahul Sharma — Documentation Responsible for project documentation, reports, and records Shiban Gurung — Frontend Development Responsible for user interface design and frontend implementation Roland Lepcha — Backend Development Responsible for server-side logic, database integration, and APIs Sarvagya Adhikari — Digital Signature Implementation Responsible for digital signature logic, certificate handling, and authentication mechanism
SUPPORT RESOURCES	The following support resources are required for successful implementation: • Supported relational database instances (e.g., PostgreSQL, MySQL) for data extraction and testing • Backend server environment to perform data processing and digital signature operations • Approved cryptographic libraries for secure digital signature generation and verification • Local development and testing infrastructure • Access to sample structured datasets for validation • Technical guidance and review support from NIC supervisory authority
SPECIAL NEEDS	• Secure access to supported relational database environments for controlled data extraction and testing • Backend environment capable of performing cryptographic operations (hashing and digital signature generation) • Access to approved digital signature keys or test certificates for implementation and validation • Secure handling mechanism for private key storage during signature operations • Controlled testing datasets representative of government data structures • Periodic technical review and validation support from NIC supervisory authority

BENEFITS AND CUSTOMERS

PROCESS OWNER	National Informatics Centre (NIC) Represented by the Project Guide / IT Director (Rajeev Rai), who is responsible for overseeing the authentication process, validating project direction, and ensuring alignment with organizational and security requirements.
KEY STAKEHOLDERS	● National Informatics Centre (NIC) ● Project Guide / IT Director – Rajeev Rai Provides direction, reviews progress, and ensures the project meets technical and security expectations. ● Project Team Members ● Rahul Sharma (Documentation) ● Shiban Gurung (Frontend) ● Roland Lepcha (Backend) ● Sarvagya Adhikari (Digital Signature) Responsible for design, development, and implementation. ● Evaluators / Review Committee Academic or organizational reviewers who assess project quality, outcomes, and compliance

FINAL CUSTOMER	Government departments and authorized organizational users who require a secure and reliable way to authenticate data using digital signatures through a web-based application.
EXPECTED BENEFITS	• Enhanced Data Integrity Ensures that imported and exported datasets remain tamper-evident and verifiable through backend digital signature mechanisms. • Improved Trust in Data Exchange Strengthens confidence in shared data across government systems by providing system-level cryptographic authenticity validation. • Risk Reduction Minimizes the possibility of unauthorized modification during data transfer or processing. • Standardized Integrity Validation Introduces a structured and repeatable approach to backend data validation within heterogeneous relational database workflows. • Support for Secure Digital Governance Aligns with government objectives for secure, reliable, and accountable data management practices.

RISKS, CONSTRAINTS, AND ASSUMPTIONS

RISKS

- Complexity in implementing backend digital signature and cryptographic mechanisms.
- Potential challenges in importing and transforming heterogeneous relational database schemas and vendor-specific data types accurately.
- Security risks related to the improper handling of digital signature keys
- Limited availability of production-level datasets for full-scale validation.
- Project timeline limitations due to internship duration.

CONSTRAINTS

- Implementation limited to internship scope and proof-of-concept level.
- No dedicated budget allocation.
- Restricted access to production government infrastructure.
- Initial implementation limited to commonly used relational database systems within available testing environment

ASSUMPTIONS

- Stable access to PostgreSQL databases for testing and validation.
- Availability of approved cryptographic libraries and digital signing mechanisms.
- Secure environment for managing digital signature keys during development.
- Availability of representative structured datasets for demonstration.
- Continued technical guidance and supervision from NIC authority

PREPARED BY	TITLE	DATE
Rahul Sharma	Student / Intern – Digital Signature Web Application Project	30/01/2026

